Solving Rational Equations

Name _____

Algebra 2 Lesson 4-5 (fast track) Date _____ Period _____

Learning goal: I can solve a rational equation by multiplying both sides by a common denominator, and I can identify any extraneous solutions.

Language goal: I can explain why a solution is or is not valid using the words rational equation, common denominator, domain and extraneous solution.

Spanish: *ecuación racional* (rational equation), *solución extraña* (extraneous solution), *dominio* (domain).

On the quiz: (1) (full Savvas quiz only): one denominator; multiply both sides by it and solve, as in Example 1. (2): add the two workers' rates, as in Practice 2. (3): multiply both sides by both denominators and solve, as in Example 1 and Practice 1. (4): clear the fractions, solve the quadratic, then cross out the **extraneous solution**. This is Example 4. (5) (full Savvas quiz only): the speed of a current is Example 5, not in this packet.

Example 1: Solve a Rational Equation

A **rational equation** contains a rational expression. Multiply both sides by the **common denominator** to eliminate the fractions, solve, then confirm the solution is valid in the original equation. What is the solution to each rational equation?

A. $\dfrac{1}{x+4} = 2$ Domain: $x \neq -4$

$$(x+4)\left(\frac{1}{x+4}\right) = 2(x+4)$$
$$1 = 2x + 8$$
$$-7 = 2x$$
$$x = -\tfrac{7}{2}$$

Check: $-\frac{7}{2} + 4 = \frac{1}{2}$ and $\frac{1}{1/2} = 2$ ✓

B. $\dfrac{1}{x-3} = 5$ Domain: $x \neq 3$

$$(x-3)\left(\frac{1}{x-3}\right) = 5(x-3)$$
$$1 = 5x - 15$$
$$16 = 5x$$
$$x = \tfrac{16}{5}$$

Check: $\frac{16}{5} - 3 = \frac{1}{5}$ and $\frac{1}{1/5} = 5$ ✓

Watch out: multiply the *whole* side by the denominator. $2(x+4)$ is $2x+8$, not $2x+4$.

Say it: I multiplied both sides by _____, the _____. The answer $x = -\frac{7}{2}$ is in the _____ because it does not make _____ equal to 0.

Try It 1: What is the solution to each equation?

a. $\dfrac{2}{x+5} = 4$ Domain: $x \neq$ _____

Solution: $x =$ _____ Check: ✓ or ×

b. $\dfrac{1}{x-7} = 2$ Domain: $x \neq$ _____

Solution: $x =$ _____ Check: ✓ or ×

Example 4: Check for Extraneous Solutions

What are the solutions to the following equations? First find the **domain**: the values that make a denominator 0 can never be solutions. A value that the cleared equation gives but the domain leaves out is an **extraneous solution**.

A. $\dfrac{5x}{x-2} = 7 + \dfrac{10}{x-2}$

Domain: $x \neq 2$.

Common denominator: $x - 2$.

$$(x-2)\left(\frac{5x}{x-2}\right) = \left(7 + \frac{10}{x-2}\right)(x-2)$$
$$5x = 7(x-2) + 10$$
$$5x = 7x - 14 + 10$$
$$-2x = -4$$
$$x = 2$$

Check: $x = 2$ makes the denominator $x - 2$ equal to 0, so 2 is an **extraneous solution**.

This equation has **no solution**.

B. $\dfrac{3}{x-3} = \dfrac{x}{x-3} - \dfrac{x}{4}$

Domain: $x \neq 3$. Common denominator: $4(x-3)$.

$$4(x-3)\left(\frac{3}{x-3}\right) = \left(\frac{x}{x-3} - \frac{x}{4}\right)4(x-3)$$
$$4(3) = 4(x) - x(x-3)$$
$$12 = 4x - x^2 + 3x$$
$$x^2 - 7x + 12 = 0 \quad \text{standard form}$$
$$(x-3)(x-4) = 0 \quad \text{factor}$$
$$x = 3 \quad \text{or} \quad x = 4$$

Check: $x = 3$ makes $x - 3 = 0$, so 3 is extraneous.

$x = 4$: $\dfrac{3}{1} = 3$ and $\dfrac{4}{1} - \dfrac{4}{4} = 4 - 1 = 3$ ✓

The only solution is $x = 4$; 3 is extraneous.

Say it: In Part B, $x = 3$ is an _____ because it makes the _____ equal to 0, so it is not in the _____ of the original equation. The only solution is _____.

Try It 4: What are the solutions to the following equations?

a. $x + \dfrac{6}{x-3} = \dfrac{2x}{x-3}$

Domain: $x \neq$ _____ Multiply by: _____

Solutions of the cleared equation: _____

Extraneous: _____ Solution: _____

b. $\dfrac{x^2}{x+5} = \dfrac{25}{x+5}$

Domain: $x \neq$ _____ Multiply by: _____

Solutions of the cleared equation: _____

Extraneous: _____ Solution: _____

In part b, the cleared equation has two solutions. Why is only one of them a solution of the original equation? Use the word *extraneous*.

Practice: Two denominators and working together

1. Two denominators (Practice 32). Which of the following is the solution of $\dfrac{3}{x+1} = \dfrac{2}{x-3}$?

A. $x = -11$ B. $x = -\frac{7}{5}$ C. $x = \frac{7}{5}$ D. $x = 11$

Domain: $x \neq$ _____ and _____ Multiply both sides by *both* denominators: _____

Solution: $x =$ _____ Check: ✓ or ×

2. Working together (Practice 26). Kenji can finish a puzzle in 2 hours working alone. Oscar can finish the same puzzle in 3 hours working alone. How long would it take Oscar and Kenji to finish the puzzle if they worked on it together?

a. Each rate is 1 puzzle divided by the hours. Kenji: _____ puzzle per hour. Oscar: _____ puzzle per hour.

b. Let t = hours working together. Their rates add up to $\frac{1}{t}$: _____ + _____ = _____

c. Multiply both sides by the common denominator _____ and solve.

d. $t =$ _____ hours

e. Your answer should be less than 2 hours. Why must the time together be less than either time alone?

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
CHP ★	Solve rational equations	Quiz 3, 4; Quiz 1 (full Savvas quiz only)
CJC ★	Solve a quadratic equation by factoring	Quiz 4: the quadratic left after clearing the fractions (Example 4B)
7JJ	Rational functions: asymptotes and excluded values	Warm-up: the excluded values are the numbers that can never be solutions
FEX	Add and subtract rational expressions	Warm-up from 4-4: finding a common denominator

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Practice 13 (DOK 3)

Higher Order Thinking. Write a rational equation that cannot have 2 or -6 as solutions.

A denominator that is 0 when $x = 2$: _____ and when $x = -6$: _____

My equation:

Why can neither 2 nor -6 be a solution of your equation? Use the words *domain* and *denominator*.

Before the quiz, finish these sentences.

To solve a rational equation, I first find the _____, then I multiply both sides by the _____.

After I solve, I check each answer. An answer that makes a denominator 0 is an _____, so I cross it out.

When two people work together, I _____ their rates, and the time together is _____ than either time alone.